NCERT · CLASS 7 · SCIENCE

Measurement of Time and Motion

C08

Measure time. Track motion.

11 NCERT questions · pendulum · speed · motion

Textbook exercise answers first · Illustrated explanations · Revision

Original learning notes based on the NCERT chapter. Diagrams are schematic and editable.

Part A · All 11 NCERT textbook exercise answers

How to use this section

Find the matching question number under *Let Us Enhance Our Learning* (textbook pp. 118–119). Each answer gives the calculation or reason, then a quick check.

Q1 · Car travels 150 m in 10 s

Given: Distance 150 m; time 10 s. $v = \dfrac{150}{10} = 15\ \text{m/s}$.

Convert with $1\ \text{m/s} = 3.6\ \text{km/h}$: $15 \times 3.6 = \boxed{54\ \text{km/h}}$.

Q2 · Two 400 m runners

Runner A: $v_A = 400/50 = 8\ \text{m/s}$. Runner B: $v_B = 400/45 = 80/9 \approx 8.89\ \text{m/s}$.

Runner B is faster by $\dfrac{80}{9} - 8 = \boxed{\frac{8}{9}\ \text{m/s} \approx 0.89\ m/s}$.

Q3 · Train: 25 m/s over 360 km

Convert distance first: $360\ \text{km} = 360000\ \text{m}$.

Time $= \dfrac{360000}{25} = 14400\ \text{s} = \boxed{4\ \text{hours}}$.

Q4 · Train covers 180 km in 3 hours

(i) $v = 180/3 = \boxed{60\ \text{km/h}}$.

(ii) $60 \times \frac{5}{18} = \boxed{\frac{50}{3}\ \text{m/s} \approx 16.67\ m/s}$.

(iii) At the same speed for 4 h: $d = 60 \times 4 = \boxed{240\ \text{km}}$.

Q5 · Horse versus train

Horse: $18\ \text{m/s} = 18 \times 3.6 = 64.8\ \text{km/h}$. Train: $72\ \text{km/h}$.

Train is faster by $\boxed{7.2\ \text{km/h}}$, equivalently $2\ \text{m/s}$.

Q6 · Uniform and non-uniform motion

A car travelling in a straight line at constant speed on an empty highway shows **uniform linear motion**: equal distances in equal intervals. In city traffic it starts, slows and stops, showing **non-uniform motion**: unequal distances in equal time intervals.

Q7 · Fill the uniform-motion table

The object moves 8 m in 10 s, so $v = 0.8\ \text{m/s}$. Thus $d = 0.8t$.

Time (s)	0	10	20	30	40	50	70
Distance (m)	0	8	16	24	32	40	56

Blanks: at 20 s the distance is 16 m; 32 m corresponds to 40 s.

Q8 · Car travels 60, 70, 50 km in successive hours

It is **not uniform**: distances in equal 1-hour intervals differ.
Average speed $= (60+70+50)/(1+1+1) = 180/3 = \boxed{60\ \text{km/h}}$.

Q9 · Which motion is common in everyday life?

Non-uniform motion is more common. For example: (1) a bus slowing at stops, (2) a bicycle speeding up or braking, and (3) a person alternately walking and resting. Speeds change with time.

Q10 · Table from 0 to 100 s

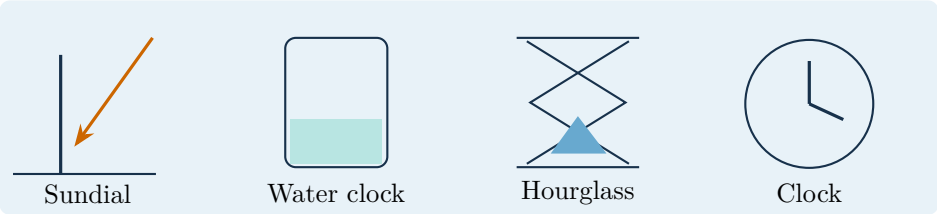
Distances at 10-s intervals are $0, 6, 10, 16, 21, 29, 35, 42, 45, 55, 60$ m. Successive increases are $6, 4, 6, 5, 8, 6, 7, 3, 10, 5$ m.
Since these increases differ, speed is **non-uniform**. Average speed $= \dfrac{60-0}{100-0} = \boxed{0.6\ \text{m/s}}$.

Q11 · Vehicle covers 2 km in 200 s

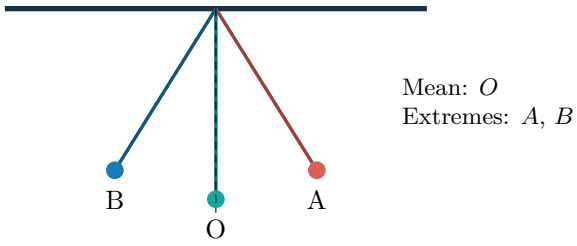
First 500 m at 10 m/s: $t_1 = 50$ s. Next 500 m at 5 m/s: $t_2 = 100$ s.
Remaining distance $= 2000 - 1000 = 1000$ m, remaining time $= 200 - 150 = 50$ s.
Required final-leg speed $= 1000/50 = \boxed{20\ \text{m/s}}$.
Whole-trip average speed $= 2000/200 = \boxed{10\ \text{m/s}}$.

Part B · Learn the chapter visually

1. Old and modern ways to measure time. Repeating natural events gave people a way to count days and seasons. Sundials use the Sun's shadow; water clocks measure flow or sinking; an hourglass uses falling sand. Today, clocks and stopwatches measure intervals more precisely.



2. Pendulum and oscillation. A simple pendulum consists of a bob hanging on a thread from a fixed support. One complete oscillation is $O \to A \to O \to B \to O$ (equivalently $A \to B \to A$). Its time period is the time taken for one full oscillation.



Activity 8.2: how to calculate a time period

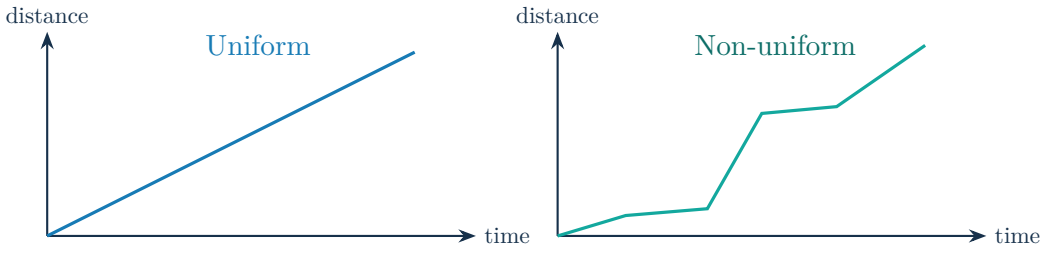
Time 10 oscillations three or four times, then compute $T =$ (time for 10 oscillations)/10. *Example data, not claimed observations:* if 10 swings take 20.4 s, then $T = 2.04$ s. Small differences in repeated readings are normal measurement errors.

3. Speed and units. Average speed $= \dfrac{\text{total distance}}{\text{total time}}$.

$$1 \text{ m/s} = 3.6 \text{ km/h}, \qquad 1 \text{ km/h} = \frac{5}{18} \text{ m/s}.$$

Do not average unequal segment speeds directly. First add the distances and the times.

4. Uniform vs non-uniform linear motion. Compare the distance covered in *equal* time intervals.



Textbook activity reminders

Train X covers 20 km in every 10-minute interval (uniform). Train Y covers unequal distances (non-uniform). In a real timetable activity, record station departure/arrival and *actual* distance before calculating speed; do not invent observations.

Common mistakes · quick fixes

- **km versus m:** convert first. 360 km is 360000 m, not 3600 m.
- **Average speed versus instantaneous speed:** average speed uses *whole journey* distance divided by total time.
- **One oscillation:** $O \to A \to O$ is *half*, not a whole cycle.
- **Non-uniform does not mean curved:** a car on a straight road may still change its speed.

Try it yourself · check your learning

1. A cyclist covers 900 m in 3 min. Find the speed in m/s.
   Answer: $900/180 = 5\ m/s$.
2. A pendulum makes 15 full oscillations in 30 s. Find the time period.
   Answer: $30/15 = 2\ s$.
3. A car travels 40 km in hour 1 and 80 km in hour 2. Uniform? Average speed?
   Answer: no; $120/2 = 60\ km/h$.

Remember: **distance ÷ time = average speed.**